

Joel Joy

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Computer scientist with a strong foundation in programming, data science, and full-stack development. Demonstrated ability to communicate complex technical concepts to non-technical stakeholders.

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, C, C++, Dart, HTML, CSS, SQL
- **Frameworks & Tools:** Claude code, Selenium, Pytorch, Flutter, TensorFlow, scikit-learn, Android Studio, Git, AWS
- **Development:** Agile, Scrum, MVVM, REST APIs, Unit Testing, Reactive Programming

PROJECTS

Topology Optimization (for U.S. Army): Developed machine learning pipeline reducing part generation time from 2 hours to 5 minutes while maintaining near-optimal designs. Generated 30,000-part training dataset using FEniCS after implementing traditional topology optimization algorithms. Led development of dcGAN and genetic algorithm approaches, evaluated against CNN-based iterative models, reinforcement learning, and simulated annealing methods.

Workhorse: Built a mobile and web app connecting employers and workers using Flutter and MongoDB. Led frontend development using Git workflows in Agile practices. Coordinated with backend developers to ensure seamless API integration and contributed to design decisions by conducting user testing sessions and feedback iterations.

NewsCheck: Built a full-stack web application that uses a fine-tuned DistilBERT model to classify the political bias of news articles on a continuous spectrum from Left to Right. Trained on 37,000+ individually labeled articles from the Article Bias Prediction dataset, achieving 86% test accuracy across three bias classes. Deployed with a Hugging Face Spaces backend and a custom frontend featuring a real-time bias spectrum visualization.

Farsight: Companion app for a real time strategy game that provides near real time advice to players about macro level decisions based on advanced machine learning models. Developed time-series model predicting jungle ganks 15 seconds in advance using 4-timestep windows with 110 features per timestep. Trained and evaluated LSTM, VAE, CVAE, and Transformer architectures on 2,500,000-row dataset (25,000 games) with 5% positive class rate, achieving 40 AP. Addressed severe class imbalance, temporal dependencies, and feature leakage through ablation studies and architectural constraints. Launching in Q1 2026.

EXPERIENCE

United Auto Insurance | QA Analyst Oct 2025 – Present

- Created and executed comprehensive test cases for insurance policy management system, handling up to 5 concurrent tickets while matching clearance rates of tenured employees.
- Interfaced directly with AS400 database to validate data integrity, diagnose system issues, and collaborate with development leads on bug isolation.
- Developed automated test framework using Selenium to improve testing efficiency.
- Led QA validation for expansion into new markets, leading to acquiring an additional 400 customers in 4 months.

Uncommon Schools | Computer Science Teacher Aug 2024 – Oct 2025

- Taught 120 high school students the fundamentals of Web Design and Python.
- Tapped by administrators to train teachers to integrate computer programming into their curriculum.
- Utilized interpersonal and mentorship skills to establish a positive student culture, as exemplified by a B+ median grade and fewer than normal disciplinary escalations.
- Lead groups of teachers to create individual programming projects for student learning.

EDUCATION

University of Central Florida (UCF)

2019 - 2023

Bachelor of Science in Computer Science